

Raemond Bergstrom-Wood

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EDUCATION

- **University of California, Berkeley** Berkeley, CA
Bachelors of Science, Electrical Engineering and Computer Science May 2014

EXPERIENCE

- **Solid Electric Vehicles** San Francisco, CA
Senior Software Engineer June 2021 - Present
 - **Firmware:**
 - ★ Bring-up, factory travel, iOS App, cloud service for multiple projects including the final C1 Bike
 - ★ Designed and managed production of the Wire harnesses
 - ★ Managed outside development of an Ambarella based camera system (3 cameras, low power wake, cloud streaming)
 - ★ Prototyped features using fingerprint unlock, radar based motion detection
 - Product: Developed the target market and positioning pitch/thesis.
 - Advanced Prototyping: Created the mechanicals (printed, cad'd and cnc'd), electronics (custom pcb), firmware and app integration for a custom lock system.
 - Brand and Web: Found and managed the branding and web development teams from concept to launch.
- **Skip Scooters** San Francisco, CA
Senior Software Engineer January 2019 - June 2021
 - **Skip S3 Firmware Lead:**
 - ★ Hardware: STM32, STM8S, Android Phone
 - ★ Interfaces: USB (Android Open Accessory), CAN, I2C, UART, RS485
 - ★ Factory/Vendor travel and build support
 - ★ Managed vendor hardware and software development
 - **Skip Ninebot Firmware Lead:**
 - ★ MCUs: PIC24F, Android Phone
 - ★ Interfaces: USB (Android Open Accessory), UART
 - ★ Inherited a legacy contractor developed project and implemented a number of simplifications and critical bug fixes
 - ★ Reduced cell data usage by 75%
 - Pushed the company to change the pricing models, from per minute to per mile. The change resulted in a 80% reduction in support contacts with no loss in revenue and increased NPS.
- **Essential Products** Palo Alto, CA
Software Engineer November 2017 - October 2018
 - **Home Accessories Firmware Lead:**
 - ★ Worked to create a home accessory road map
 - ★ Developed FreeRTOS based app and diag application
 - ★ Wrote sensor, wireless, and hardware interface libraries
 - ★ Designed and implemented ultrasonic sensor algorithms & filtering
 - ★ Brought up and developed the signal processing algorithms for 24GHz Radar based Human presence detection
 - ★ Worked with chip vendors through SOW/MSA to develop custom chip firmware
 - ★ CM Selection and Bringup
 - ★ Worked with a third party contractor to get through a POC electrical (schematics/layout) and PD (mechanical) design.
- **Arete Labs, a Stealth Home Security Company (Acquired by Essential)** San Francisco, CA
Co-Founder and only Software Engineer August 2016 - November 2017
 - **Firmware:**
 - ★ Developed FreeRTOS based app and diag application
 - ★ Wrote sensor, wireless, service connection and hardware interface libraries
 - ★ Designed and implemented sensor fusion & filtering
 - ★ WiFi Soft AP based pairing

- ★ Device/Service in app pairing/provisioning
- ★ Push notifications
- Developed and designed a feature complete iOS app developed in Swift.
- Industrial design, UI, UX ideation and feedback
- Maintained system block diagram and reviewed all schematics & layouts
- LT SPICE 120VAC Circuit Modeling & Analog Design
- Patents: US20190120471A1, WO2019018317A1

• Nest Labs/Google

RF Hardware & Software Engineer (Intern from May 2013 - June 2014)

Palo Alto, CA
May 2013 - August 2016

- **Nest Guard & Sense:** RF Hardware Lead
 - ★ LTE, WCDMA, GSM, WiFi, BLE, Thread (802.15.4) integration into Nest Secure
 - ★ Debugged and found manufacturable solutions for desensed radios (and sensors)
 - ★ Impedance matching filters, FEMS, LNAs, PAs
 - ★ Schematic and Layout Reviews
 - ★ Drove CM and In-house Validation
 - ★ Factory build support
 - ★ FCC, UL Certification Pretesting and Preparation
- **Nest Protect:**
 - ★ Led validation of the Nest Protect alarm propagation over 802.15.4 in a crowded 2.4GHz environment
 - ★ Wrote python tool to automate all RF validation testing from the physical layer (integrating everything from RS232 power supplies, vector signal analyzers and spectrum analyzers) to networking applications
- Supported numerous projects in development, debugging RF circuits and modules, interacting directly with CMs, embedded platform and communication software teams
- Interviewed all new team members for RF Hardware Design Group and the Factory Software Team
- Experience with UL, FCC, ETSI, BT Sig Certification Testing
- Created/Architect-ed Nest's factory test system and brought up/hired a team to support it
- Built an internal website to host and plot RF validation data (Python, Flask, GAE)
- Patent: US10381722B2